Safe&Sound: Wearable Environmental Sound Alerts for Hearing-Impaired Users

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***Abstract*—People with hearing impairments often miss critical environmental cues like smoke alarms, sirens, or doorbells that healthy hearing enables. The inability to detect such sounds can compromise safety and independence, which makes this a significant public and social challenge. Existing solutions (such as smartphone or smartwatch sound-notification services) offer helpful assistance but tend to rely on cloud processing or continuous connectivity, and they typically don't transfer well into lightweight wearable devices. Many research prototypes target broad audio classification tasks but don't constrain themselves to microcontroller hardware or optimize for power, latency, and privacy. Our project, Safe&Sound, closes that gap by deploying a quantized TinyML model on an ESP32 to perform local, real-time classification of emergency environmental sounds. A companion ESP32 wristband receives alerts over BLE and provides color-coded LED and haptic feedback. The system supports offline-first operation, user personalization of alerts, and secure, optional cloud updates.**

***Index Terms*—IoT, TinyML, Environmental Sound Classification, Accessibility, ESP32, BLE, Wearable Devices**

I. Introduction

While technology continues to expand the boundaries of human capability, it has not yet fully bridged the gap for individuals with sensory impairments. One group that remains particularly underserved is the hearing-impaired community, which faces safety and accessibility challenges in daily life due to their inability to detect environmental sounds such as alarms, sirens, or doorbells. According to the World Health Organization, over 430 million people currently experience disabling hearing loss, a number projected to grow to nearly 700 million by 2050. The inability to perceive auditory warnings directly impacts safety, independence, and overall quality of life, particularly in situations involving fires, home intrusions, or approaching vehicles.

To address these challenges, significant research has been conducted in environmental sound recognition (ESR) and assistive wearable technology. Deep learning methods have shown considerable promise for sound classification, with convolutional and recurrent networks achieving high accuracy. Gairí et al. present an extensive review of ESR on embedded devices, noting that while deep models can perform well, their deployment on microcontrollers remains constrained by memory, power, and real-time processing limitations [1]. Similarly, Ting et al. introduce an Inception-Dense architecture optimized for hearing aids, achieving strong noise classification results with reduced computation and latency [3]. These works demonstrate that accurate real-time audio inference is feasible, yet they primarily focus on algorithmic efficiency rather than user-centric assistive applications.

On the applied side, researchers have explored various wearable and IoT-based solutions for sound awareness. Jain et al. developed SoundWatch, a smartwatch application that uses lightweight CNN models to alert deaf and hard-of-hearing users to everyday sounds, achieving up to 97% accuracy on high-priority alerts [4]. Zhu et al. designed an edge-computing wearable that recognizes emergency vehicle sirens with over 95% accuracy using an EfficientNet-based ensemble running entirely on a microcontroller [2]. Earlier work by Yağanoğlu and Köse also demonstrated the practical value of vibration-based feedback wearables for hearing-impaired individuals, reporting 98–99% accuracy in identifying common household and outdoor alert sounds [5]. Collectively, these studies affirm the societal relevance and technical feasibility of sound-aware assistive systems.

Despite these advances, several gaps persist in the current body of work. Many systems are designed around a single sound class (e.g., sirens or alarms) rather than a generalized multi-class emergency-sound framework. Others rely heavily on smartphones or cloud connectivity, introducing latency, privacy, and dependency concerns [1, 2]. Furthermore, few designs integrate TinyML-level model deployment with customizable haptic and visual feedback under the strict memory and energy constraints of microcontrollers. Finally, empirical evaluations often omit practical hardware metrics such as inference latency, energy consumption, and end-to-end user response time which are essential for a fully deployable assistive IoT solution.

To close these gaps, this paper proposes Safe&Sound, a two-device Level-4 IoT system that merges TinyML-based sound classification with real-time wearable feedback. The first device, an ESP32 detector, captures audio through an I²S microphone and performs on-device inference using a quantized convolutional neural network trained via transfer learning. The second device, an ESP32 wristband, receives Bluetooth Low Energy (BLE) notifications and provides immediate color-coded LED and vibration alerts. The system supports offline-first operation for privacy and reliability, while

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a companion mobile application enables cloud-synchronized configuration—allowing users to personalize alert mappings, vibration patterns, and detection thresholds.

Our proposed solution contributes to society by enabling accessible, privacy-preserving sound awareness for individuals with hearing impairments. Also, by integrating wearable IoT communication and user-customizable feedback in a compact, low-power form factor. Finally, by providing a framework that can extend to other safety-critical domains, such as elderly care or smart-home monitoring. Through Safe&Sound, we aim to advance the inclusivity and practicality of assistive auditory technology, bringing real-time environmental awareness to those who need it most.

II. System Design

A. Process Model

This process model represents the real-time decision flow of Safe&Sound. It captures how environmental sounds are continuously analyzed, classified, and acted upon. The simplicity of this design ensures low latency, clear logic, and repeatable behavior.

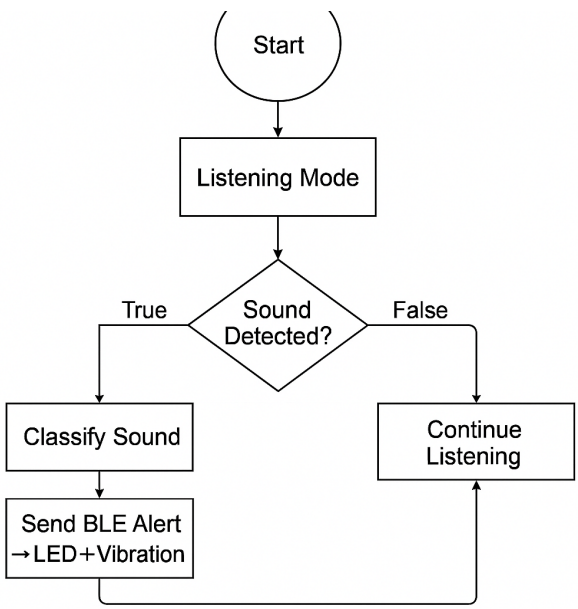

Fig. 1. Process Model Specification for Safe&Sound.

B. Domain Model Specification

This domain model demonstrates how Safe&Sound integrates sensing, processing, communication, and feedback into a cohesive assistive IoT framework. It emphasizes the direct mapping between real-world stimuli of sounds and their digital interpretations of classified alerts, ensuring timely and personalized user awareness.

C. Information Model Specification

The information model specification defines the structure of data managed within the Safe/Sound system. Each detected sound and corresponding alert is represented as a virtual entity with defined state attributes that describe the system's behavior. The Detected Sound entity captures the classification and confidence level of an incoming audio signal, while the Alert entity represents the resulting notification transmitted to the wristband. These attributes determine whether a sound is identified as an emergency, doorbell, or other event, and define the visual and haptic feedback patterns used to alert the user. Together, these virtual entities outline how physical environmental sounds are translated into meaningful, actionable states within the IoT system.

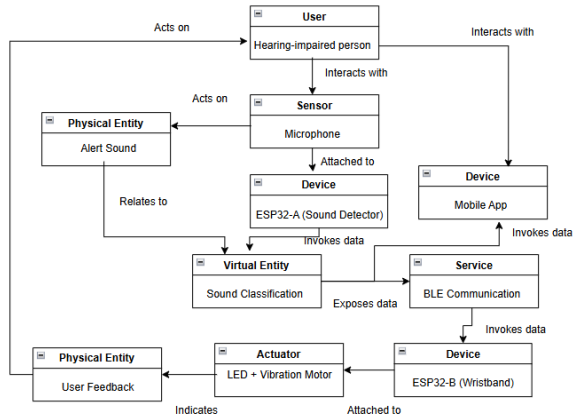

Fig. 2. Domain Model Specification for Safe&Sound.

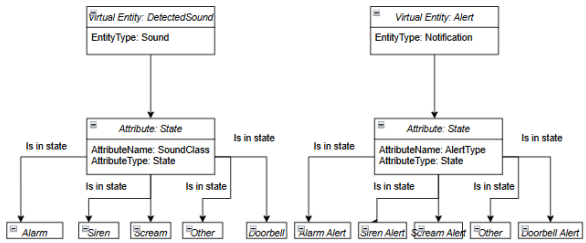

Fig. 3. Information Model Specification for Safe&Sound.

D. Service Specification

The service specification details the functional communication between system components, including the capture, classification, and notification of critical sounds. The input audio from the microphone serves as the sensing element, processed through the TinyML-based classification service on the ESP32-A, and transmitted over Bluetooth Low Energy (BLE) to the wristband. The BLE service acts as the system's primary communication layer, ensuring real-time data transfer and low-power operation. The output of this service drives the wristband's LED and vibration responses, providing immediate feedback to the hearing-impaired user.

E. IoT Level Specification

The IoT level specification illustrates the deployment hierarchy of the Safe&Sound system across both the local and cloud

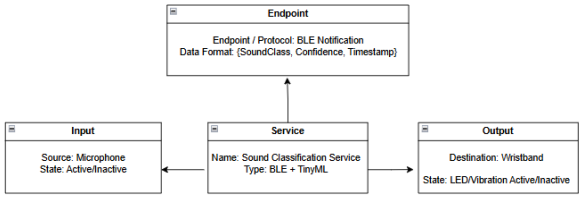

Fig. 4. Service Specification for Safe&Sound.

environments. The local layer consists of two ESP32 devices, one functioning as the controller node for audio sensing and TinyML classification, and the other serving as an actuator wristband that provides LED and vibration feedback through BLE communication. The cloud layer, comprising the Flutter mobile application and Firebase Firestore, handles real-time data logging, user configuration, and event visualization.

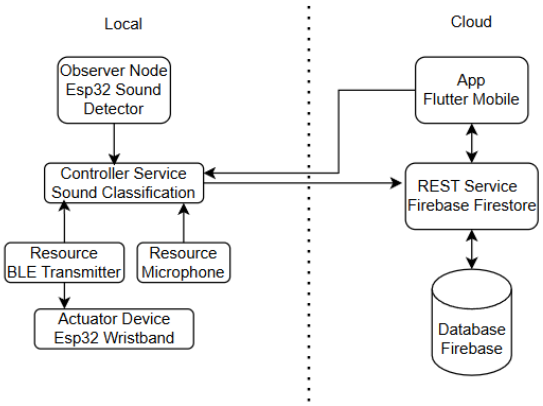

Fig. 5. IoT Level Specification for Safe&Sound.

F. Functional View Specification

The functional view specification illustrates how the Safe&Sound system integrates its hardware and software components. The device layer consists of ESP32-based nodes for sound detection and feedback, equipped with a microphone, BLE module, LED, and vibration motor. These devices communicate through BLE and Wi-Fi protocols managed by the communication and service modules. Firebase Firestore and the Flutter mobile application form the core of the application layer, providing cloud-based storage, visualization, and user interaction. Management and security modules ensure reliable device operation and protect user data, creating a secure, cloud-connected IoT architecture for assisting hearing-impaired users.

G. Real

The real-world setup consists of two primary components: an ESP32-based sound detector and a wearable wristband. The sound detector captures and classifies environmental sounds using a digital microphone and TinyML model running locally

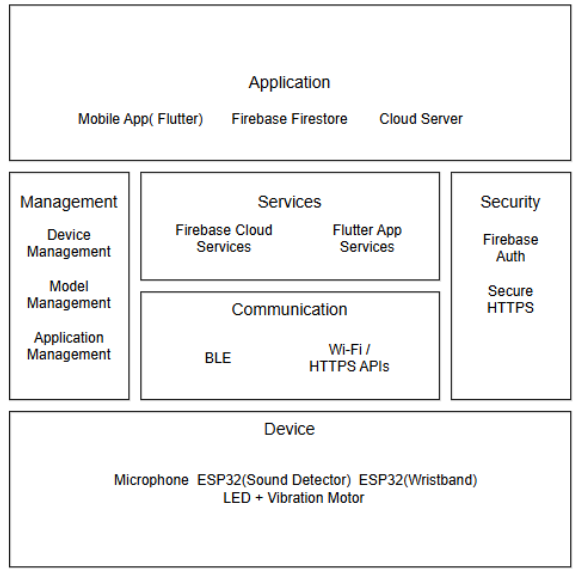

Fig. 6. Functional View Specification for Safe&Sound.

on the ESP32. When a critical sound is identified it transmits an alert wirelessly to the wristband via Bluetooth. The wristband provides intuitive feedback to the user through an RGB LED that changes color based on the detected sound type and a vibration motor that ensures tactile notification for hearing-impaired users. This design emphasizes simplicity, portability, and real-time awareness.

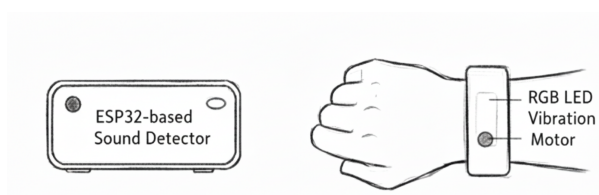

Fig. 7. Physical Sketch for Safe&Sound.

H. Metrics

To evaluate system performance, multiple quantitative and qualitative metrics will be used. Detection accuracy will be measured as the percentage of correctly classified emergency sounds across multiple trials in varied noise environments. Latency will be measured from the moment a sound occurs to the time the wristband displays an alert, targeting sub-500 ms response. BLE communication reliability and range will be assessed by testing packet loss and signal stability at increasing distances. Power efficiency will be measured by tracking runtime on battery power for both the sound detector and wristband devices.

I. Plan for Testing

The system will be demonstrated in a controlled indoor environment replicating household conditions. A set of recorded sounds including fire alarms, doorbells, shouts, and glass

breaking will be played through a speaker at varying distances. The ESP32 detector will classify each event using the on-device TinyML model and transmit results via BLE to the wristband. Observers will verify that the correct LED color and vibration pattern activate for each sound type. The Flutter app will display real-time alerts and event logs retrieved from Firebase. Final testing will include range validation, battery life measurement, and noise interference analysis to demonstrate robustness and portability.

J. Target User

The target users of Safe&Sound are individuals who are deaf or hard of hearing and may miss important auditory cues in their surroundings. In a typical use case, a user wears the wristband while at home or on the go. When a critical sound occurs such as a smoke alarm, siren, or doorbell, the ESP32 detector classifies it and sends a BLE alert to the wristband, which vibrates and flashes a corresponding color pattern. The user can then check the app for visual confirmation and context. This provides independence and safety assurance for users who cannot rely on auditory warnings alone.

K. Security Considerations

ince the system captures and classifies audio, privacy and data security are key design concerns. Audio is processed entirely on-device, and only classification labels, not raw sound, are transmitted or stored, minimizing risk of data exposure. BLE communication between devices will use pairing-based encryption, and data sent to Firebase will use HTTPS with authenticated app access. The app will restrict user data to device IDs and alert logs only, ensuring no personally identifiable or sensitive information is stored. These measures maintain user privacy while still enabling reliable, connected functionality.

III. Bill of Materials (BOM)

Level	Part Number	Part Name	Description	Quantity	Cost per Part ($)
1	0-100	ESP32-WROVER	Microcontroller	2	15.00
2	0-101	INMP441 I2S Microphone	Digital microphone for sound detection	1	5.50
2	0-102	DiCUNO RGB LED	Visual indicator for alerts	1	2.50
2	0-103	Coin Vibration Motor (3V)	Haptic feedback actuator	1	2.00
2	0-104	LiPo Battery (3.7V 1000 mAh)	Power source for both devices	2	6.00
2	0-105	TP4056 Charger Module (USB-C)	Battery charging and protection	2	2.00
2	0-106	Slide Power Switch (SPDT)	Power control switch for each device	2	0.50
2	0-107	Breadboard and Jumper Wires	Prototyping and wiring components	1	8.00
2	0-108	PLA Filament (100 g)	3D printing material for enclosures	1	2.50
2	0-109	Velcro Strap Material	Adjustable wristband strap	1	2.00
1	0-110	TensorFlow Lite Micro	On-device TinyML inference library	1	0.00
1	0-111	Flutter SDK	Mobile app development framework	1	0.00
1	0-112	Firebase (Free Tier)	Cloud backend and database	1	0.00

TABLE I
Bill of Materials for Safe&Sound System

References

[1] G. Gairí, J. Riba, and X. Giró-i-Nieto, "Environmental sound recognition on embedded devices using deep learning: a review," *Artificial Intelligence Review*, vol. 58, 2025.

[2] C. Zhu, L. Xu, and Y. Yang, "A Wearable Assistant Device for the Hearing Impaired to Recognize Emergency Vehicle Sirens with Edge Computing," *Sensors*, vol. 23, no. 15, pp. 583–596, 2023.

[3] K. Ting, H. Tan, and M. L. Tay, "Environmental Noise Classification with Inception-Dense Blocks for Hearing Aids," *Sensors*, vol. 21, no. 9, pp. 505–514, 2021.

[4] A. Jain, N. Patel, and J. Froehlich, "SoundWatch: Exploring Smartwatch-Based Deep Learning Approaches to Support Sound Awareness for Deaf and Hard of Hearing Users," in *Proc. ACM ASSETS*, 2020, pp. 1–14.

[5] V. Yağanoğlu and H. Köse, "Real-Time Detection of Important Sounds with a Wearable Vibration-Based Device for Hearing-Impaired People," *Electronics*, vol. 7, no. 8, pp. 482–490, 2018.